

Lesson 5-1: Determine the Area of Parallelograms and Rhombuses

Connect & apply • 8 tasks • Prepares for Standard 6.G.A.1

Name: _____ Date: _____ Period: _____

Show complete thinking. Label units. Use an equation, sketch, or model. Accuracy before speed.

TASK 1 • FOUNDATION 1

Compute 24×13 .

TASK 2 • FOUNDATION 2

A rectangle is 8 cm long and 6 cm wide. Find its area in square centimeters.

TASK 3 • FOUNDATION 3

A parallelogram has base 10 cm, slanted side 7 cm, and perpendicular height 4 cm. Which number is the height?

TASK 4 • FOUNDATION 4

A rectangle measures 3 meters by 5 meters. State its area with units.

One strategy or model I used: _____

My next step: ☐ Explain my steps ☐ Try another problem ☐ Review with a partner

Lesson 5-1: Determine the Area of Parallelograms and Rhombuses

Connect & apply • 8 tasks • Prepares for Standard 6.G.A.1

Name: _____ Date: _____ Period: _____

Show complete thinking. Label units. Use an equation, sketch, or model. Accuracy before speed.

TASK 5 • QUICK CHECK A

Area of an 8 cm by 5 cm rectangle? In a parallelogram, is the height the slanted side?

TASK 6 • QUICK CHECK B

Area of a 9 cm by 7 cm rectangle? In a parallelogram, is the height the slanted side?

TASK 7 • QUICK CHECK C

Area of a 12 m by 4 m rectangle? A parallelogram has base 12 m, slant side 5 m, height 4 m. What is its area?

TASK 8 • TRANSFER & EXTEND

Two parallelograms both have base 10 cm and height 6 cm but different slant sides. Compare their areas.

One strategy or model I used: _____

My next step: ☐ Explain my steps ☐ Try another problem ☐ Review with a partner

Lesson 5-2: Determine the Area of Triangles

Connect & apply · 8 tasks · Prepares for Standard 6.G.A.1

Name: _____ Date: _____ Period: _____

Show complete thinking. Label units. Use an equation, sketch, or model. Accuracy before speed.

TASK 1 • FOUNDATION 1

Compute $(14 \times 9) \div 2$.

TASK 2 · FOUNDATION 2

Compute $\frac{1}{2} \times 38$.

TASK 3 • FOUNDATION 3

A parallelogram has base 12 cm and height 5 cm.
Find its area in square centimeters.

TASK 4 · FOUNDATION 4

A triangle has base 8 cm, perpendicular height 6 cm, and slanted side 10 cm. Which length belongs in the height part of its area formula?

One strategy or model I used: _____

My next step: ☐ Explain my steps ☐ Try another problem ☐ Review with a partner

Lesson 5-2: Determine the Area of Triangles

Connect & apply • 8 tasks • Prepares for Standard 6.G.A.1

Name: _____ Date: _____ Period: _____

Show complete thinking. Label units. Use an equation, sketch, or model. Accuracy before speed.

TASK 5 • QUICK CHECK A

7×4 , then half? Area of a triangle with base 7 m and height 4 m?

TASK 6 • QUICK CHECK B

9×6 , then half? Area of a triangle with base 9 cm and height 6 cm?

TASK 7 • QUICK CHECK C

12×5 , then half? Area of a triangle with base 12 in and height 5 in?

TASK 8 • TRANSFER & EXTEND

A triangle has area 36 cm^2 and base 12 cm. What is its height?

One strategy or model I used: _____

My next step: ☐ Explain my steps ☐ Try another problem ☐ Review with a partner

Lesson 5-3: Determine the Area of Trapezoids

Connect & apply • 8 tasks • Prepares for Standard 6.G.A.1

Name: _____ Date: _____ Period: _____

Show complete thinking. Label units. Use an equation, sketch, or model. Accuracy before speed.

TASK 1 • FOUNDATION 1

Compute $(7 + 11) \times 4$.

TASK 2 • FOUNDATION 2

Compute $1/2 \times 54$.

TASK 3 • FOUNDATION 3

Evaluate $1/2 \times (6 + 10) \times 5$.

TASK 4 • FOUNDATION 4

A shape is made of a rectangle of area 24 and two triangles of area 6 each, without overlap. Find the total area.

One strategy or model I used: _____

My next step: ☐ Explain my steps ☐ Try another problem ☐ Review with a partner

Lesson 5-3: Determine the Area of Trapezoids

Connect & apply • 8 tasks • Prepares for Standard 6.G.A.1

Name: _____ Date: _____ Period: _____

Show complete thinking. Label units. Use an equation, sketch, or model. Accuracy before speed.

TASK 5 • QUICK CHECK A

$\frac{1}{2} \times (6 + 10) \times 4 = ?$ Do the parentheses first.

TASK 6 • QUICK CHECK B

$\frac{1}{2} \times (8 + 12) \times 5 = ?$ Do the parentheses first.

TASK 7 • QUICK CHECK C

$\frac{1}{2} \times (7 + 9) \times 6 = ?$ Do the parentheses first.

TASK 8 • TRANSFER & EXTEND

A trapezoid has bases 6 and 10 and area 48.
What is its height?

One strategy or model I used: _____

My next step: ☐ Explain my steps ☐ Try another problem ☐ Review with a partner

Lesson 5-4: Apply Area Concepts to Solve Problems

Connect & apply · 8 tasks · Prepares for Standard 6.G.A.1

Name: _____ Date: _____ Period: _____

Show complete thinking. Label units. Use an equation, sketch, or model. Accuracy before speed.

TASK 1 • FOUNDATION 1

Find the area of a triangle with base 10 cm and height 8 cm, in square centimeters.

TASK 2 · FOUNDATION 2

A rectangle has area 72 square meters. A 15-square-meter section is removed. What area remains?

TASK 3 • FOUNDATION 3

A total side length is 14 cm. One part is 9 cm.
How long is the other part?

TASK 4 • FOUNDATION 4

Six identical tiles each cover 12 square centimeters. What is the total area?

One strategy or model I used: _____

My next step: ☐ Explain my steps ☐ Try another problem ☐ Review with a partner

Lesson 5-4: Apply Area Concepts to Solve Problems

Connect & apply • 8 tasks • Prepares for Standard 6.G.A.1

Name: _____ Date: _____ Period: _____

Show complete thinking. Label units. Use an equation, sketch, or model. Accuracy before speed.

TASK 5 • QUICK CHECK A

A 10 by 6 rectangle has a 3 by 2 rectangle cut out. What area is left?

TASK 6 • QUICK CHECK B

A 12 by 8 rectangle has a 4 by 3 rectangle cut out. What area is left?

TASK 7 • QUICK CHECK C

A 9 by 9 square has a triangle of base 6 and height 4 cut out. What area is left?

TASK 8 • TRANSFER & EXTEND

A rectangular garden 15 m by 10 m has a 3 m by 3 m square shed inside. How much garden is left to plant?

One strategy or model I used: _____

My next step: ☐ Explain my steps ☐ Try another problem ☐ Review with a partner

Lesson 5-5: Determine the Volume of Rectangular Prisms

Connect & apply · 8 tasks · Prepares for Standard 6.G.A.2

Name: _____ Date: _____ Period: _____

Show complete thinking. Label units. Use an equation, sketch, or model. Accuracy before speed.

TASK 1 • FOUNDATION 1

Compute $4 \times 7 \times 5$.

TASK 2 · FOUNDATION 2

Compute $1\frac{1}{2} \times \frac{2}{3}$.

TASK 3 • FOUNDATION 3

A rectangular base is 9 cm by 4 cm. Find its area in square centimeters.

TASK 4 · FOUNDATION 4

A prism measures 2 cm by 3 cm by 5 cm. State its volume with units.

One strategy or model I used: _____

My next step: ☐ Explain my steps ☐ Try another problem ☐ Review with a partner

Lesson 5-5: Determine the Volume of Rectangular Prisms

Connect & apply · 8 tasks · Prepares for Standard 6.G.A.2

Name: _____ Date: _____ Period: _____

Show complete thinking. Label units. Use an equation, sketch, or model. Accuracy before speed.

TASK 5 • QUICK CHECK A

$4 \times 3 \times 2 \frac{1}{2} \times \frac{1}{3}$? Volume of a 4 by 3 by 2 prism?

TASK 6 • QUICK CHECK B

5 × 4 × 3? 1/2 × 1/4? Volume of a 5 by 4 by 3 prism?

TASK 7 • QUICK CHECK C

$6 \times 2 \times 4 \times \frac{2}{3} \times \frac{1}{2}$? Volume of a 6 by 2 by 4 prism?

TASK 8 · TRANSFER & EXTEND

A box is $3\frac{1}{2}$ in by 2 in by 4 in. What is its volume?

One strategy or model I used: _____

My next step: ☐ Explain my steps ☐ Try another problem ☐ Review with a partner

Lesson 5-6: Represent Three-Dimensional Figures in Two Dimensions

Connect & apply · 8 tasks · Prepares for Standard 6.G.A.4

Name: _____ Date: _____ Period: _____

Show complete thinking. Label units. Use an equation, sketch, or model. Accuracy before speed.

TASK 1 • FOUNDATION 1

How many faces, edges, and vertices does a rectangular prism have?

TASK 2 · FOUNDATION 2

How many faces does a square pyramid have, including its base?

TASK 3 • FOUNDATION 3

A net has a rectangular face 3 cm by 7 cm. Find that face's area in square centimeters.

TASK 4 • FOUNDATION 4

A prism has face pairs sized 3 by 4, 3 by 5, and 4 by 5. How many faces are 3 by 4?

One strategy or model I used: _____

My next step: ☐ Explain my steps ☐ Try another problem ☐ Review with a partner

Lesson 5-6: Represent Three-Dimensional Figures in Two Dimensions

Connect & apply • 8 tasks • Prepares for Standard 6.G.A.4

Name: _____ Date: _____ Period: _____

Show complete thinking. Label units. Use an equation, sketch, or model. Accuracy before speed.

TASK 5 • QUICK CHECK A

How many faces does a rectangular prism have?
How many matching pairs is that?

TASK 6 • QUICK CHECK B

How many faces does a triangular prism have?
How many are triangles?

TASK 7 • QUICK CHECK C

How many faces does a square pyramid have?
How many are triangles?

TASK 8 • TRANSFER & EXTEND

A net has two congruent right triangles with side lengths 3, 4, and 5 units, and three rectangles measuring 3 by 10, 4 by 10, and 5 by 10 units. The triangle legs of lengths 3 and 4 are perpendicular. What solid does the net form, and what is its surface area?

One strategy or model I used: _____

My next step: ☐ Explain my steps ☐ Try another problem ☐ Review with a partner

Lesson 5-7: Determine Surface Area of Prisms

Connect & apply · 8 tasks · Prepares for Standard 6.G.A.4

Name: _____ Date: _____ Period: _____

Show complete thinking. Label units. Use an equation, sketch, or model. Accuracy before speed.

TASK 1 · FOUNDATION 1

Add the face areas $12 + 12 + 15 + 15 + 20 + 20$.

TASK 2 · FOUNDATION 2

Find the area of a triangle with base 6 cm and height 9 cm, in square centimeters.

TASK 3 · FOUNDATION 3

In a rectangular prism with dimensions 2, 3, 5, list the three different face areas.

TASK 4 · FOUNDATION 4

One face has area 18 square centimeters. What is the combined area of it and its matching opposite face?

One strategy or model I used: _____

My next step: ☐ Explain my steps ☐ Try another problem ☐ Review with a partner

Lesson 5-7: Determine Surface Area of Prisms

Connect & apply • 8 tasks • Prepares for Standard 6.G.A.4

Name: _____ Date: _____ Period: _____

Show complete thinking. Label units. Use an equation, sketch, or model. Accuracy before speed.

TASK 5 • QUICK CHECK A

Faces measure 12, 12, 20, 20, 15, 15 square units
— what is the surface area?

TASK 6 • QUICK CHECK B

Faces measure 18, 18, 24, 24, 30, 30 square units
— what is the surface area?

TASK 7 • QUICK CHECK C

A prism has faces 10, 10, 14, 14, 35, 35 square units
— what is the surface area?

TASK 8 • TRANSFER & EXTEND

A cube has surface area 150 cm². What is the length of one edge?

One strategy or model I used: _____

My next step: ☐ Explain my steps ☐ Try another problem ☐ Review with a partner

Lesson 5-8: Determine Surface Area of Pyramids

Connect & apply · 8 tasks · Prepares for Standard 6.G.A.4

Name: _____ Date: _____ Period: _____

Show complete thinking. Label units. Use an equation, sketch, or model. Accuracy before speed.

TASK 1 · FOUNDATION 1

Find the area of a triangle with base 8 cm and perpendicular height 7 cm, in square centimeters.

TASK 2 · FOUNDATION 2

Find the area of a square base with side 6 cm, in square centimeters.

TASK 3 · FOUNDATION 3

Four congruent triangular faces each have area 21 square centimeters. Find their combined area.

TASK 4 · FOUNDATION 4

A pyramid's vertical height is 9 cm. The altitude drawn within a triangular face is 10 cm. Which height finds that face's area?

One strategy or model I used: _____

My next step: ☐ Explain my steps ☐ Try another problem ☐ Review with a partner

Lesson 5-8: Determine Surface Area of Pyramids

Connect & apply • 8 tasks • Prepares for Standard 6.G.A.4

Name: _____ Date: _____ Period: _____

Show complete thinking. Label units. Use an equation, sketch, or model. Accuracy before speed.

TASK 5 • QUICK CHECK A

One triangular face is 24 square inches. With four identical faces, what is the lateral area?

TASK 6 • QUICK CHECK B

One triangular face is 30 square inches. With four identical faces, what is the lateral area?

TASK 7 • QUICK CHECK C

A square pyramid has four congruent triangular faces, each with area 18 cm^2 , and a square base with area 36 cm^2 . What is its total surface area?

TASK 8 • TRANSFER & EXTEND

A square pyramid has base edge 10 cm and slant height 12 cm. Find the total surface area.

One strategy or model I used: _____

My next step: ☐ Explain my steps ☐ Try another problem ☐ Review with a partner